RIG-SMOOTH IS A LOCAL PROPERTY ON ARROWS OF THE CATEGORY OF ADIC NOETHERIAN TOPOLOGICAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a preliminary base-change identification of completed naive cotangent complexes, followed by verification of the three locality conditions, with the descent step done by faithful flatness through the cited Ext base-change identity inside the delivered span

Editorial note. The label lemma-rig-smooth-axioms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The proof of condition (2) applies the cited base-change identity with the author's parenthetical '(minor details omitted)'; the identity itself is a cited standard lemma (delivered as a navigation pointer), and the surrounding annihilation and faithful flatness arguments are complete in the delivered span.

Allowed standard prerequisites (editorial).

- The category WAdm^{Noeth} of adic Noetherian topological rings with continuous morphisms, and the local-property criterion of Formal Spaces for arrows
- The completed naive cotangent complex with its functorial exact sequence, as developed in the same chapter
- Base change and descent behaviour of the Ext annihilation characterizing rig-smoothness, from More on Algebra

Source reference. Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Prerequisite pointer invoked at proof line(s) 4348; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00U9: sources/stacks-chow/tags.txt, source lines 1103–1103. Prerequisite pointer invoked at proof line(s) 4311; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 4298; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0ANB: sources/stacks-chow/tags.txt, source lines 13434–13434. Prerequisite pointer invoked at proof line(s) 4278; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN9: sources/stacks-chow/tags.txt, source lines 13432–13432. Prerequisite pointer invoked at proof line(s) 4359; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0ANI: sources/stacks-chow/tags.txt, source lines 13441–13441. Prerequisite pointer invoked at proof line(s) 4256; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAF: sources/stacks-restricted/pinned-original-source.tex, source lines 560–560. Prerequisite pointer invoked at proof line(s) 4302; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 4303; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at proof line(s) 4287; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07D4: sources/stacks-restricted/pinned-original-source.tex, source lines 780–780. Prerequisite pointer invoked at proof line(s) 4314; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A6A: sources/stacks-chow/tags.txt, source lines 12838–12838. Prerequisite pointer invoked at proof line(s) 4355; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9H: sources/stacks-chow/tags.txt, source lines 20302–20302. Prerequisite pointer invoked at proof line(s) 4335; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 4334; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The property $P(\varphi) =$"φ is rig-smooth" on arrows of $WAdm^{Noeth}$ is a local property as defined in Formal Spaces, Remark 0ANI.*

Proof. Let us recall what the statement signifies. First, $WAdm^{Noeth}$ is the category whose objects are adic Noetherian topological rings and whose morphisms are continuous ring homomorphisms. Consider a commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & (B')^\wedge \\ {\scriptstyle\varphi}\uparrow & & \uparrow{\scriptstyle\varphi'} \\ A & \longrightarrow & (A')^\wedge \end{array}$$

satisfying the following conditions: A and B are adic Noetherian topological rings, $A \to A'$ and $B \to B'$ are étale ring maps, $(A')^\wedge = \lim A'/I^nA'$ for some ideal of definition $I \subset A$, $(B')^\wedge = \lim B'/J^nB'$ for some ideal of definition $J \subset B$, and $\varphi : A \to B$ and $\varphi' : (A')^\wedge \to (B')^\wedge$ are continuous. Note that $(A')^\wedge$ and $(B')^\wedge$ are adic Noetherian topological rings by Formal Spaces, Lemma 0ANB. We have to show

(1) φ is rig-smooth $\Rightarrow \varphi'$ is rig-smooth,
(2) if $B \to B'$ faithfully flat, then φ' is rig-smooth $\Rightarrow \varphi$ is rig-smooth, and
(3) if $A \to B_i$ is rig-smooth for $i = 1, \ldots, n$, then $A \to \prod_{i=1,\ldots,n} B_i$ is rig-smooth.

The equivalent conditions of Lemma 04U5 satisfy conditions (1), (2), and (3). Thus in verifying (1), (2), and (3) for the property "rig-smooth" we may already assume our ring maps satisfy the equivalent conditions of Lemma 04U5 in each case.

Pick an ideal of definition $I \subset A$. By the remarks above the topology on each ring in the diagram is the I-adic topology and B, $(A')^\wedge$, and $(B')^\wedge$ are in the category (0AL4) for (A, I). Since $A \to A'$ and $B \to B'$ are étale the complexes $NL_{A'/A}$ and $NL_{B'/B}$ are zero and hence $NL^\wedge_{(A')^\wedge/A}$ and $NL^\wedge_{(B')^\wedge/B}$ are zero by Lemma 0GAF. Applying Lemma 00S2 to $A \to (A')^\wedge \to (B')^\wedge$ we get isomorphisms

$$H^i(NL^\wedge_{(B')^\wedge/(A')^\wedge}) \to H^i(NL^\wedge_{(B')^\wedge/A})$$

Thus $NL^\wedge_{(B')^\wedge/A} \to NL_{(B')^\wedge/(A')^\wedge}$ is a quasi-isomorphism. The ring maps $B/I^nB \to B'/I^nB'$ are étale and hence are local complete intersections (Algebra, Lemma 00U9). Hence we may apply Lemmas 00S2 and 07D4 to $A \to B \to (B')^\wedge$ and we get isomorphisms

$$H^i(NL^\wedge_{B/A} \otimes_B (B')^\wedge) \to H^i(NL^\wedge_{(B')^\wedge/A})$$

We conclude that $NL^\wedge_{B/A} \otimes_B (B')^\wedge \to NL^\wedge_{(B')^\wedge/A}$ is a quasi-isomorphism. Combining these two observations we obtain that

$$NL^\wedge_{(B')^\wedge/(A')^\wedge} \cong NL^\wedge_{B/A} \otimes_B (B')^\wedge$$

in $D((B')^\wedge)$. With these preparations out of the way we can start the actual proof.

Proof of (1). Assume φ is rig-smooth. Then there exists a $c \geq 0$ such that $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ is annihilated by I^c for every B-module N. By More on Algebra, Lemmas 0G9G and 0G9H this property is preserved under base change by $B \to (B')^\wedge$. Hence $\mathrm{Ext}^1_{(B')^\wedge}(NL^\wedge_{(B')^\wedge/(A')^\wedge}, N)$ is annihilated by $I^c(A')^\wedge$ for all $(B')^\wedge$-modules N which tells us that φ' is rig-smooth. This proves (1).

To prove (2) assume $B \to B'$ is faithfully flat and that φ' is rig-smooth. Then there exists a $c \geq 0$ such that $\mathrm{Ext}^1_{(B')^\wedge}(NL^\wedge_{(B')^\wedge/(A')^\wedge}, N')$ is annihilated by $I^c(B')^\wedge$ for every $(B')^\wedge$-module N'. The composition $B \to B' \to (B')^\wedge$ is flat (Algebra, Lemma 00MB) hence for any B-module N we have

$$\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N) \otimes_B (B')^\wedge = \mathrm{Ext}^1_{(B')^\wedge}(NL^\wedge_{B/A} \otimes_B (B')^\wedge, N \otimes_B (B')^\wedge)$$

by More on Algebra, Lemma 0A6A part (3) (minor details omitted). Thus we see that this module is annihilated by I^c. However, $B \to (B')^\wedge$ is actually faithfully flat by our assumption that $B \to B'$ is faithfully flat (Formal Spaces, Lemma 0AN9). Thus we conclude that $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ is annihilated by I^c. Hence φ is rig-smooth. This proves (2).

To prove (3), setting $B = \prod_{i=1,\ldots,n} B_i$ we just observe that $NL^\wedge_{B/A}$ is the direct sum of the complexes $NL^\wedge_{B_i/A}$ viewed as complexes of B-modules. $\square$